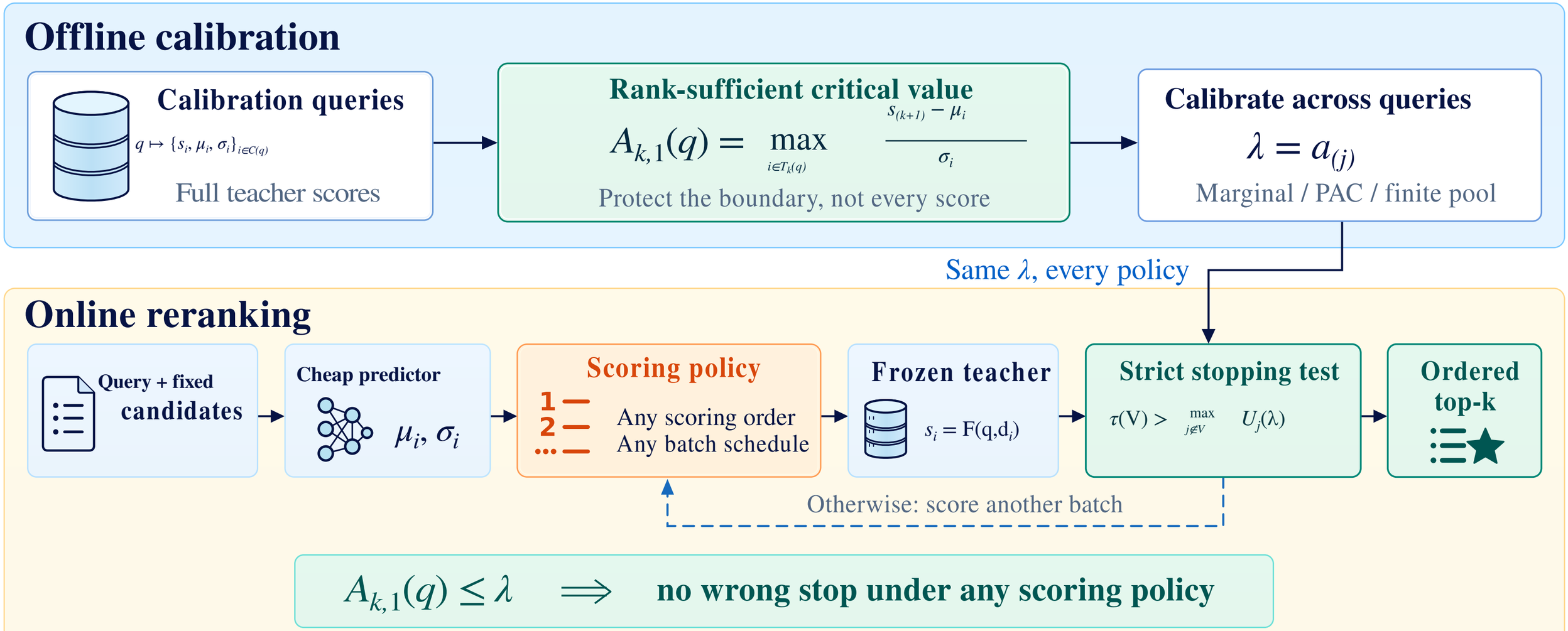
Offline calibration
Calibration queries
$q \mapsto \{s_i, \mu_i, \sigma_i\}_{i \in C(q)}$
Full teacher scores
Rank-sufficient critical value
$A_{k,1}(q) = \max_{i \in T_k(q)} \frac{s_{(k+1)} - \mu_i}{\sigma_i}$
Protect the boundary, not every score
Calibrate across queries
$\lambda = a_{(j)}$
Marginal / PAC / finite pool
Same λ, every policy
Online reranking
Query + fixed candidates
Cheap predictor
μ_i, σ_i
Scoring policy
1
2
...
Any scoring order
Any batch schedule
Frozen teacher
$s_i = F(q,d_i)$
Strict stopping test
$\tau(V) > \max_{j \notin V} U_j(\lambda)$
Ordered top-k
Otherwise: score another batch
$A_{k,1}(q) \le \lambda \implies$ no wrong stop under any scoring policy